zweitstimme.org

Forecasting the German Federal Election 2025 — Different Modelling Approaches

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The Zweitstimme.org Project

The Team Behind Zweitstimme.org


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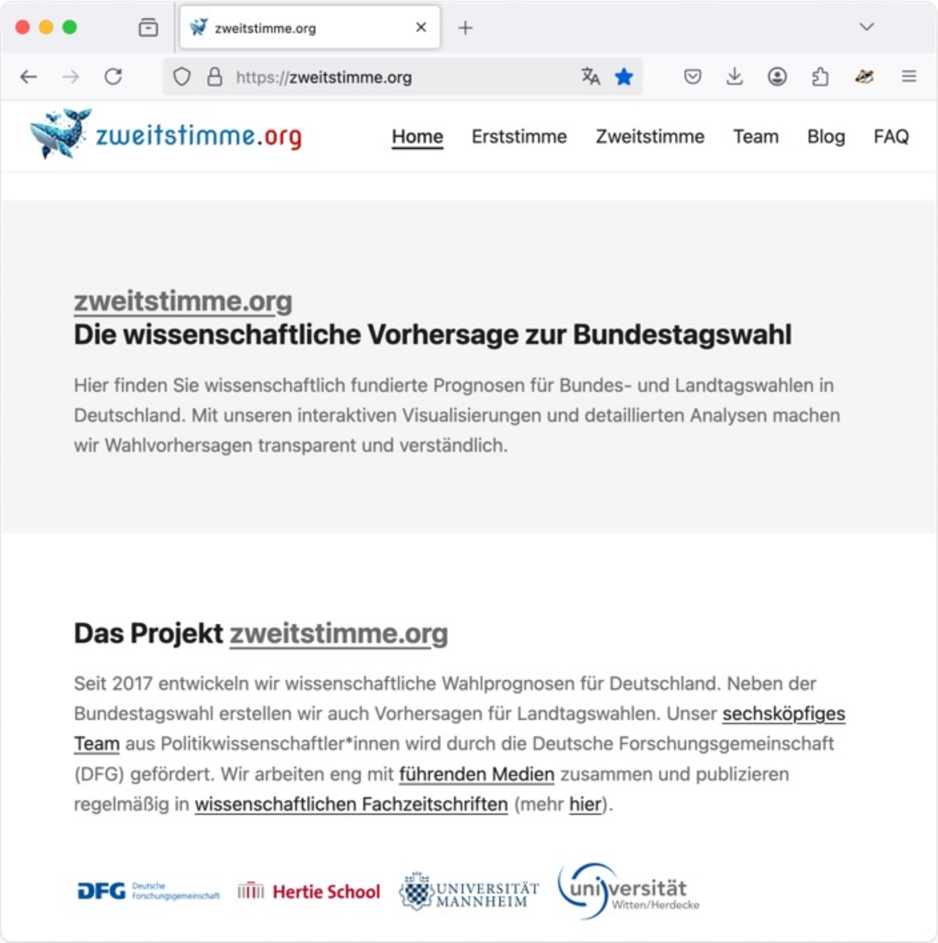

Website Traf c

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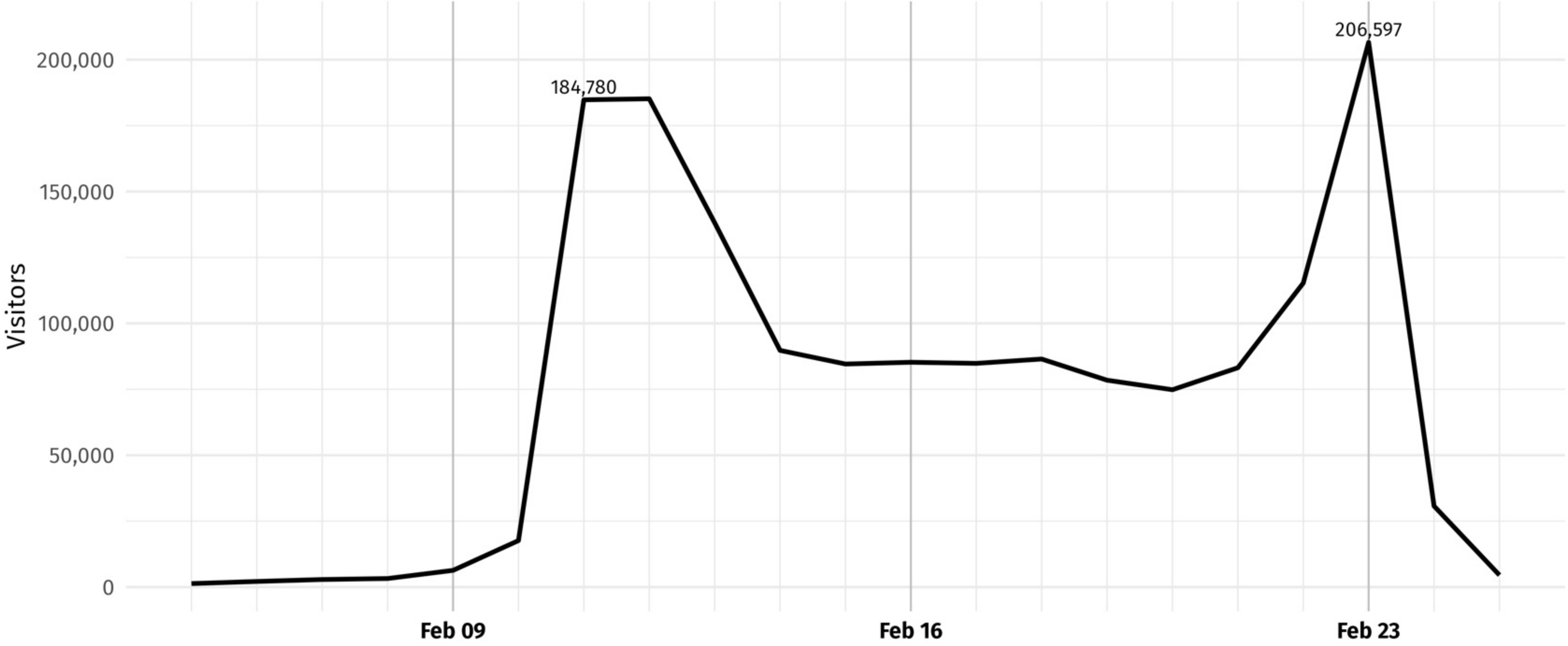
Development of visitor numbers on Zweitstimme.org

A total of 1,566,183 unique visitors



Source: Estimated unique visitors

In the Media

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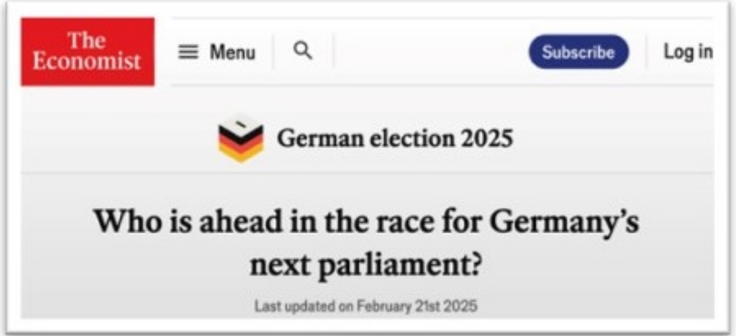
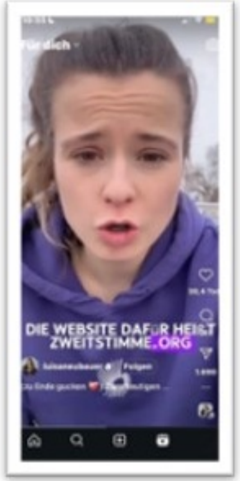
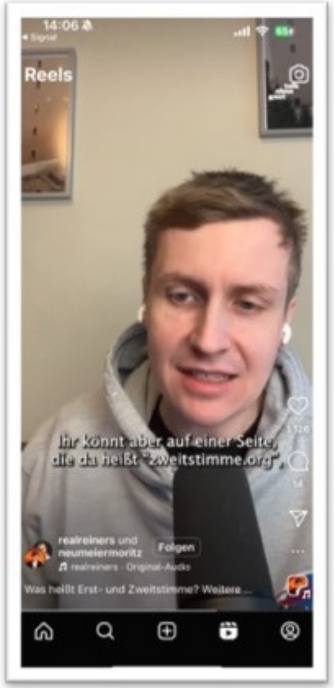
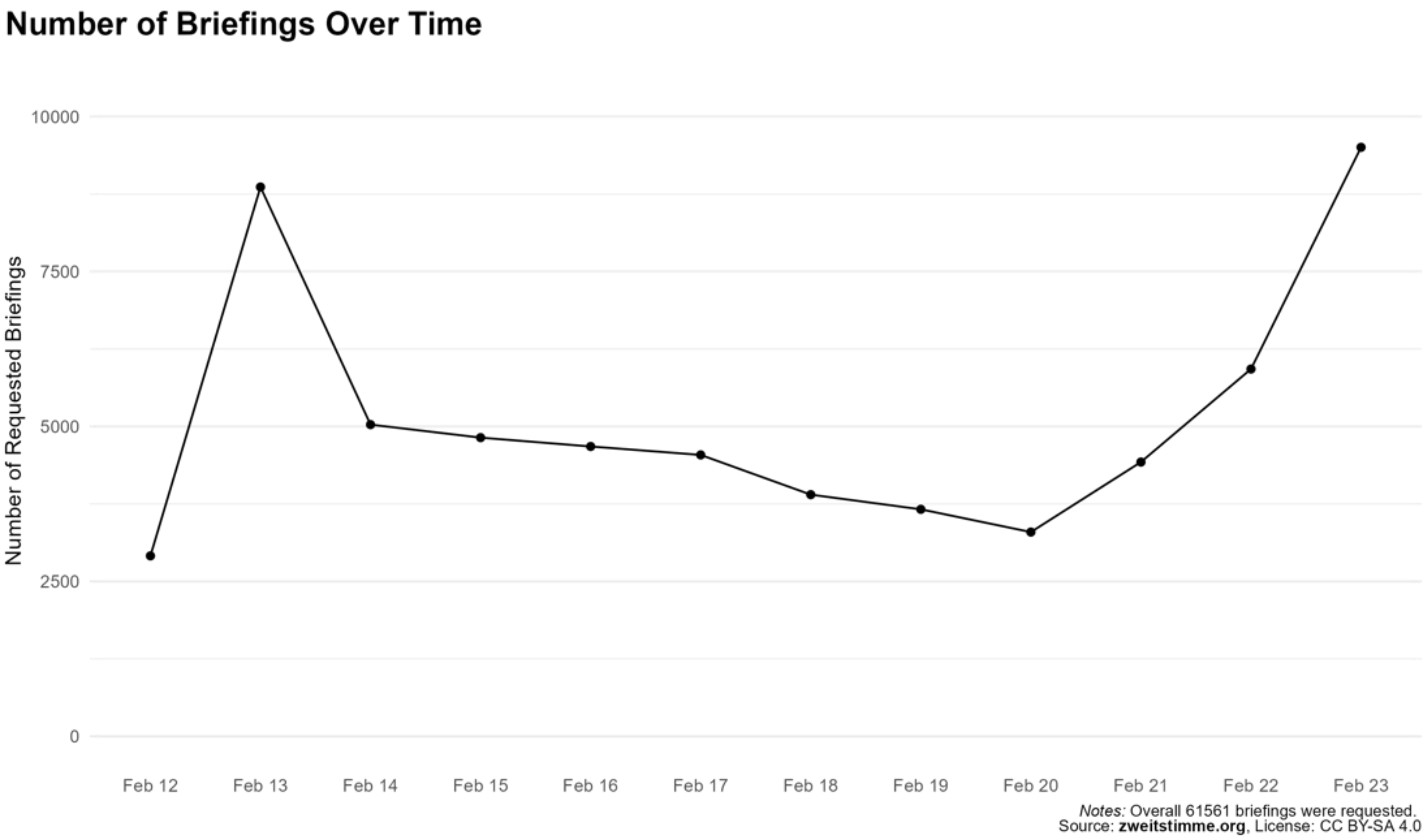
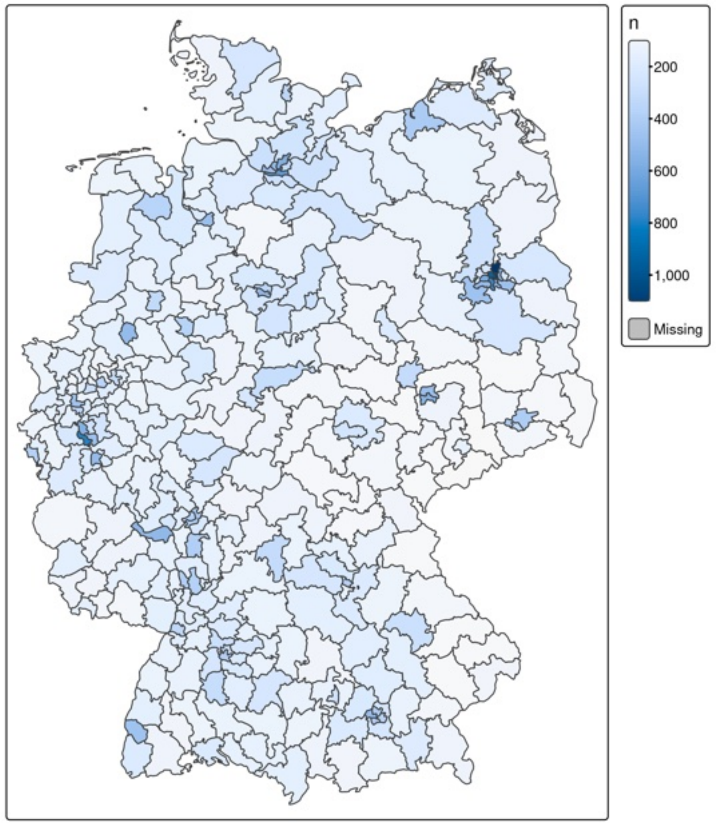






Interest in Constituency Brie ngs

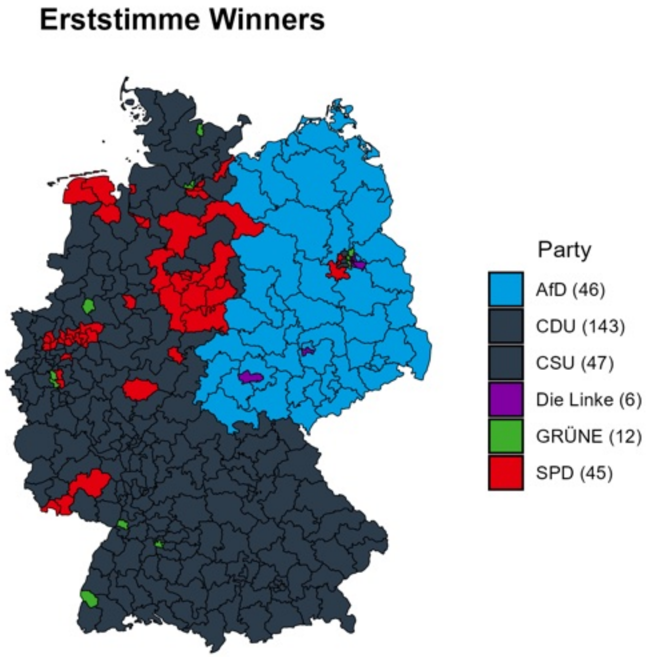




The German Electoral System

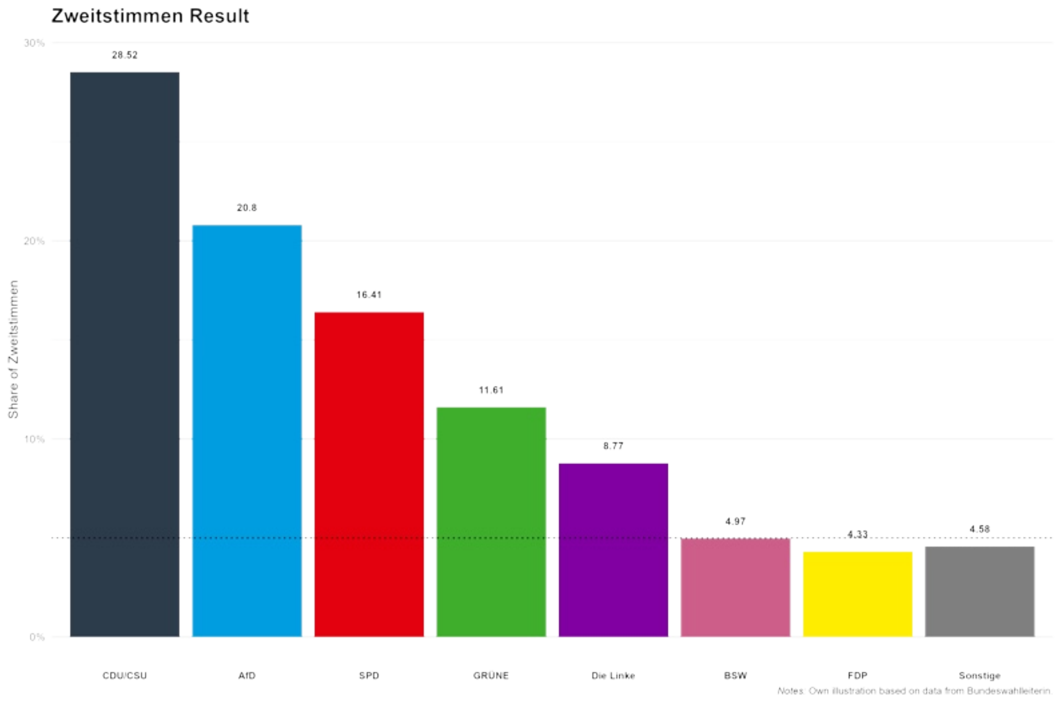
Erststimme

(" rst vote")



Zweitstimme

("second vote")



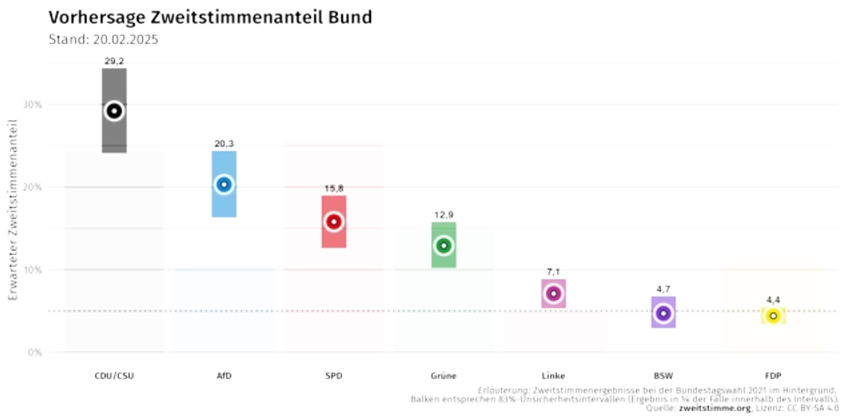
Our Forecasts

Zweitstimme Model

- Dynamic model
- "Pooling the polls"
- Data: Published polls, election results
- Implemented since 2017

→ Erststimme Model

- Zweitstimme Model + Constituency Model

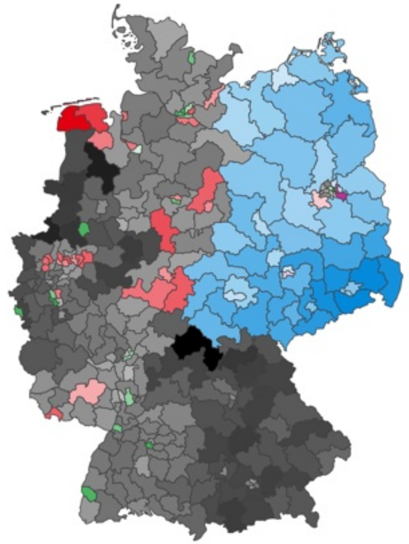


Citizen Forecast

- Survey of 50k citizens (Jan–Feb 2025)
- Asked for expectations on various election outcomes

→ incl. Erststimme and Zweitstimme

- Aggregate individual expectations into forecasts (wisdom of crowds)



Zweitstimme Forecast

The Zweitstimme Model & Citizen Forecasting

The Zweitstimme Model in Detail

Fundamentals Component

- Long-term party af liation: Results from previous election
- Short-term campaign dynamics: Average voting intentions 230-200 days before election day
- Indicator for the chancellor's party

Dynamic Component

- Public opinion polls
- Continuous data collection
- Consideration of polling institutes

Model Characteristics

- Dependent on available polling data
- Continuous data collection required
- Consideration of new parties
- Communication of uncertainties

References

- Erfort et al. (2025)
- Munzert et al. (2022)
- Gschwend et al. (2021)
- Stoetzer et al. (2019)

The Polls

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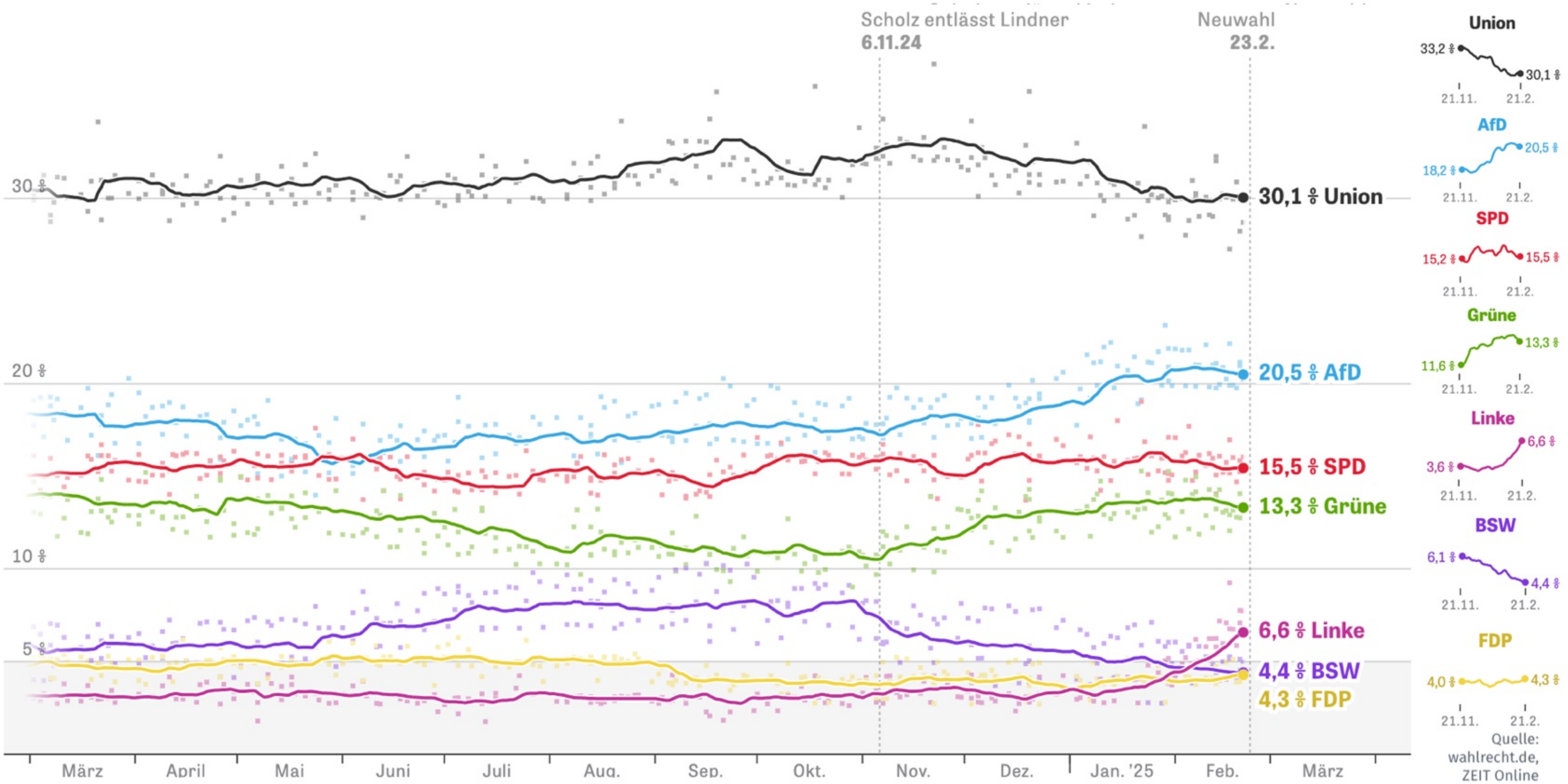

Model Forecast: Zweitstimme Federal, Dynamics

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CDU/CSU AfD SPD Greens FDP BSW Left

Evaluation: Zweitstimme Model

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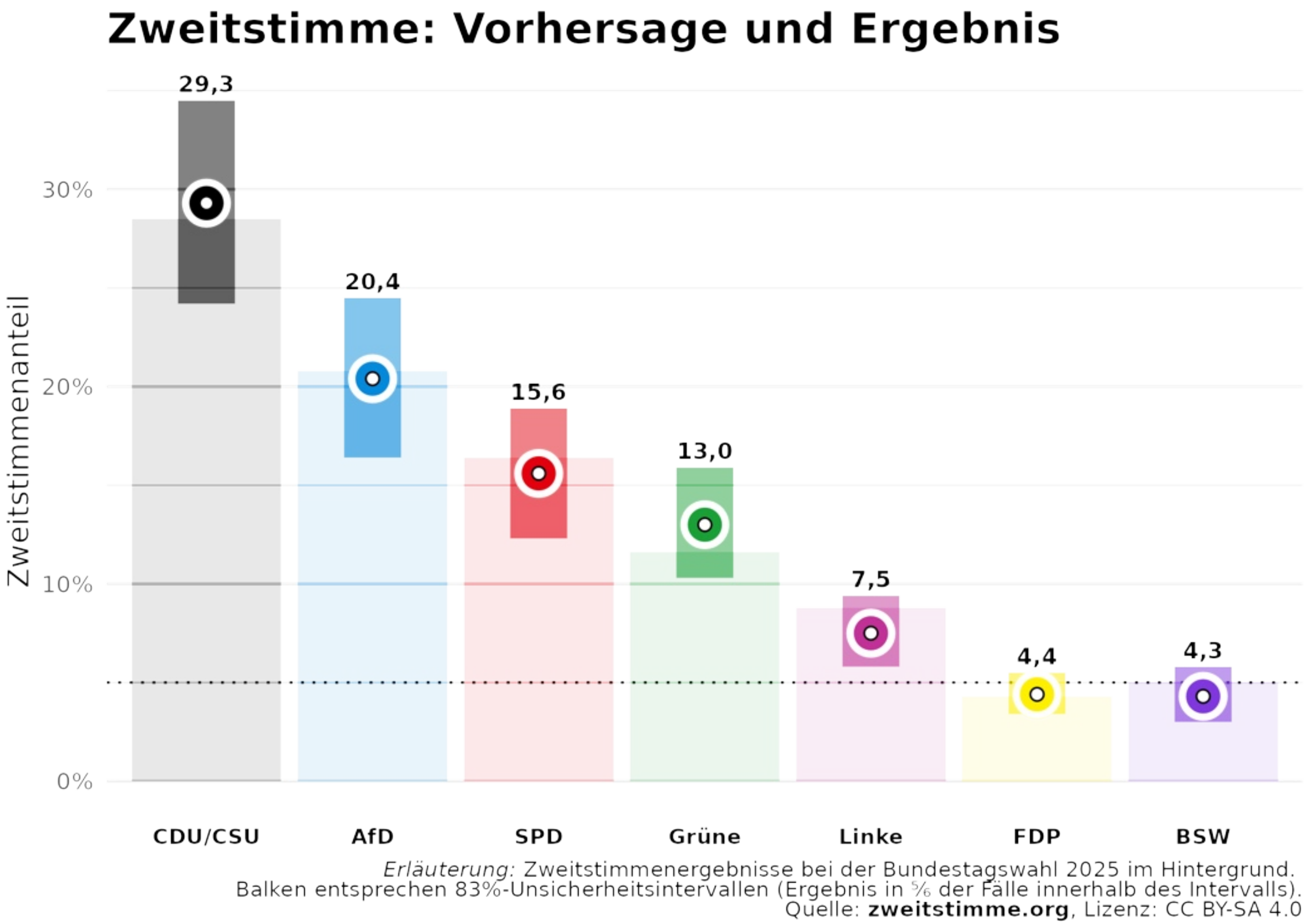
Model Performance

- Average absolute deviation: 0.78 percentage points
- All parties within predicted uncertainty intervals
- Strong performance in coalition predictions



Citizen Forecasting

Core idea: Asking individuals for their expectations on an election outcome and aggregating them into a citizen forecast.

Previous Applications

- First application in the US (Lewis-Beck & Skalaban, 1989)
- Applications in Germany (Ganser & Riordan, 2015; Graefe, 2016; Lehrer et al., 2019; Leininger et al., 2025)

Our Survey

- Over 50k Germans eligible to vote (online access panel & social media ads)
- 90-400 respondents in each of the 299 districts

Expectation Questions

- Zweitstimmen shares for each party
- Erststimmen winner in respondent's district
- Erststimmen shares for each party in district
- Likelihood of parties entering parliament
- Coalition & Chancellor

Aggregating Expectations into Forecast

- Mean (vote shares)
- Mode

Citizen Forecast: From Zweitstimme Expectations to Forecasts

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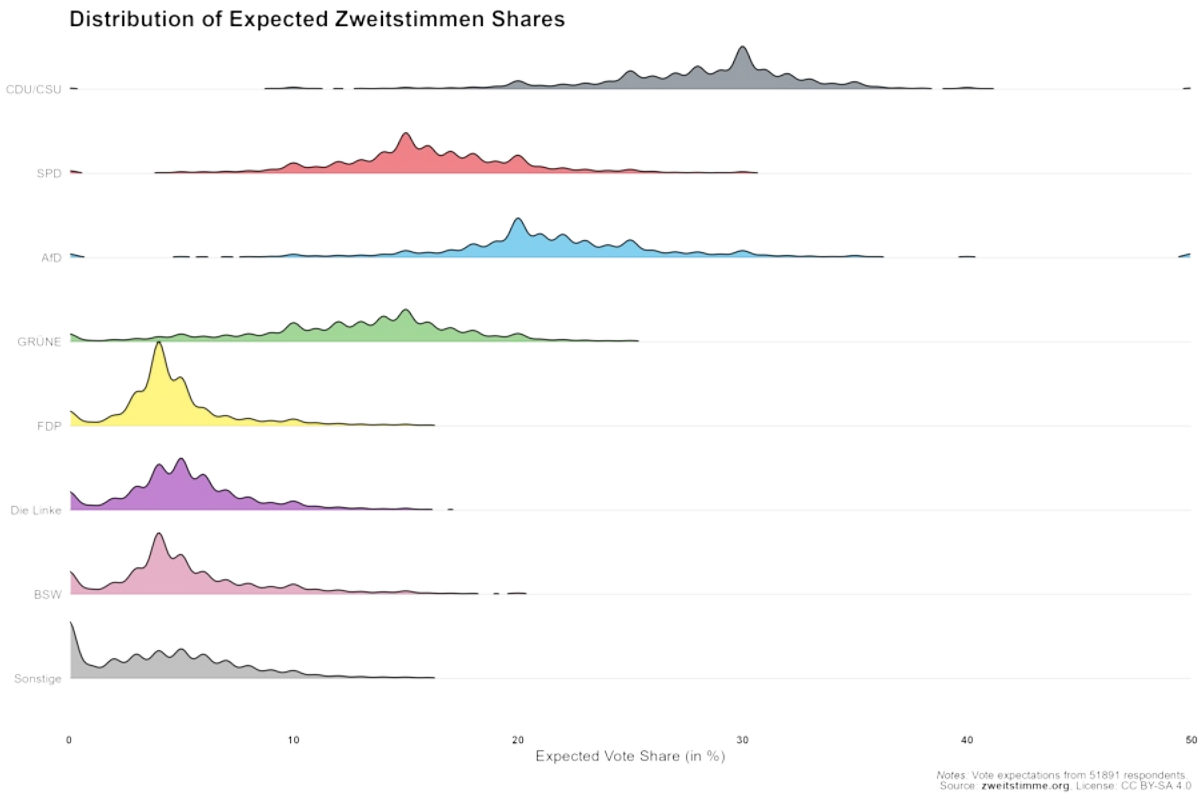
What percentage of the votes do you expect the various parties to receive nationwide?



Citizen Forecast: From Zweitstimme Expectations to Forecasts

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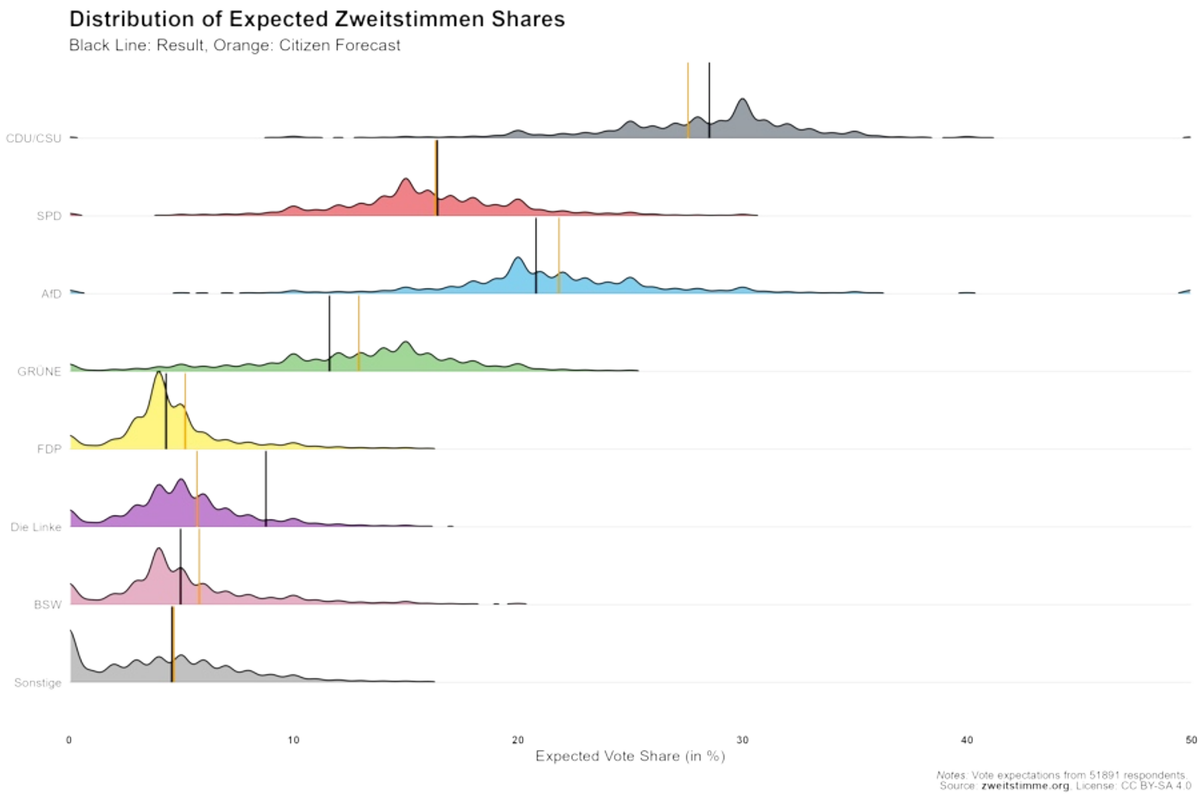

Citizen Forecast: Zweitstimme vs. Result

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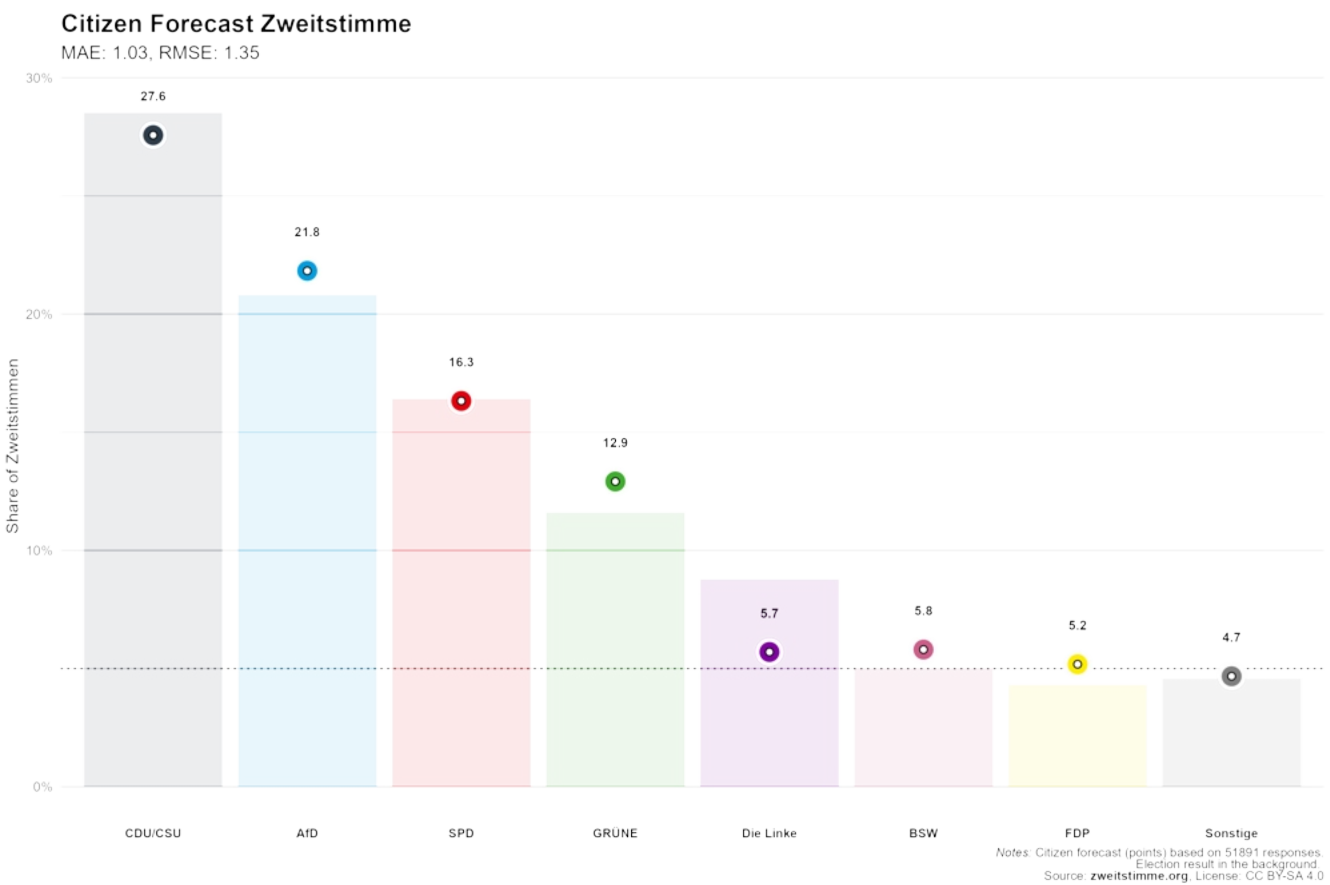
Model Performance: Average absolute deviation of 1.03 percentage points



Citizen Forecast: The Wisdom of the Crowd

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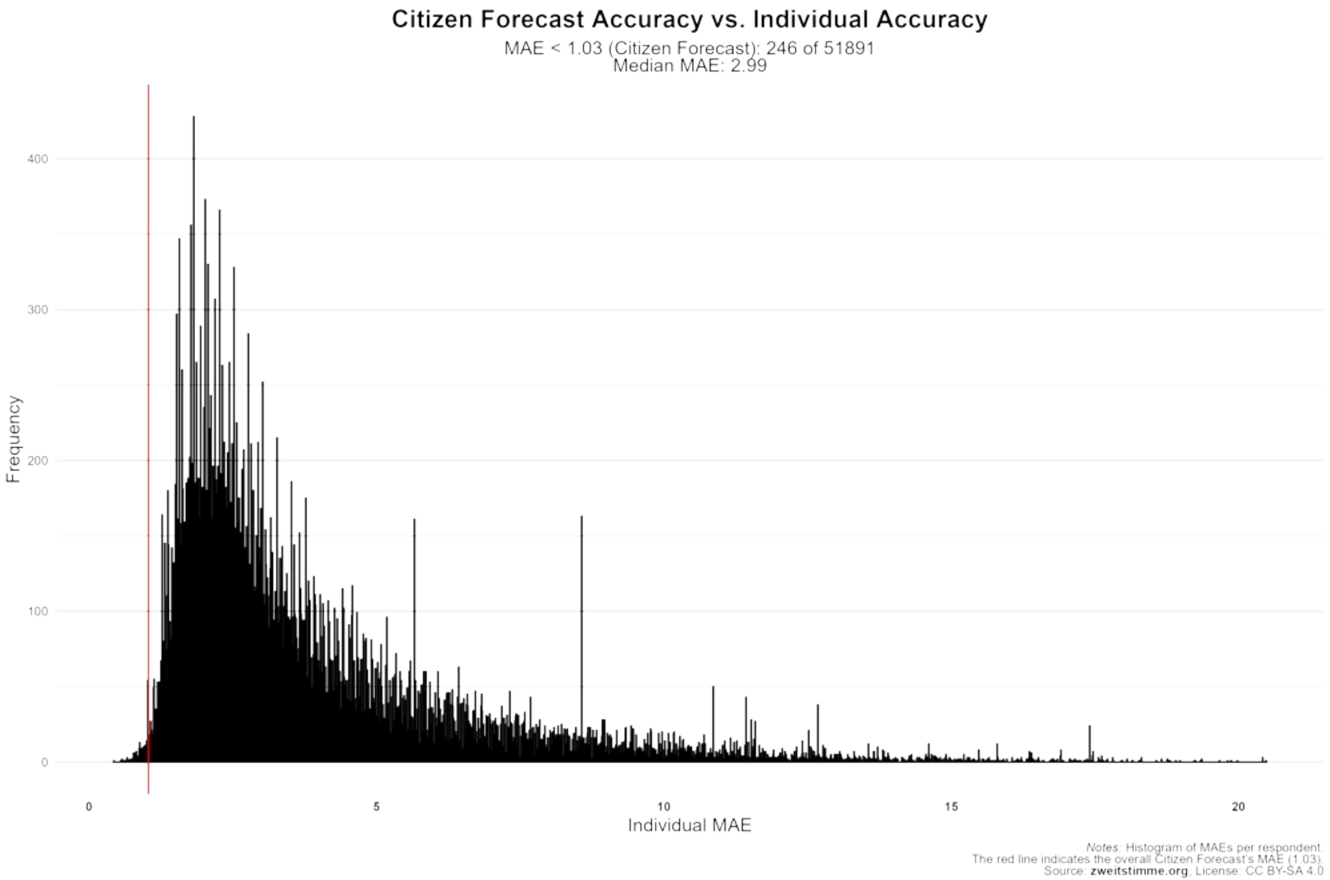
Only 0.5% of the individuals were more accurate than the group.



Erststimme Forecast

The Erststimme Model & Citizen Forecasting

The Erststimme Model in Detail

Model Foundation

- Based on Zweitstimme model predictions
- Proportional swing approach for district-level estimates
- First implemented for 2017 election
- Updated for 2021 and 2025 elections

Model Updates

- Adjusted proportional change assumptions for Left Party and BSW

District-Level Factors

- Previous election's Erststimme and Zweitstimme results
- Number of candidates in district
- East German district indicator
- Incumbent candidate indicator

Candidate-Level Factors

- Relative list position
- Presence on state list
- Previous candidacy experience
- Gender indicator
- Incumbency status
- Academic title (Dr.)
- Party's previous candidate status

Results: Erststimme

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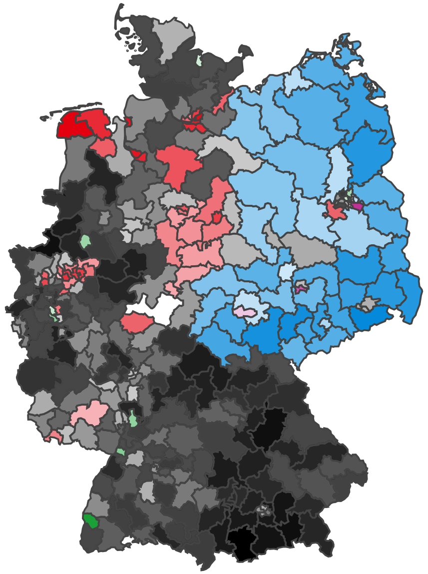
Model Forecast
Erststimme Share

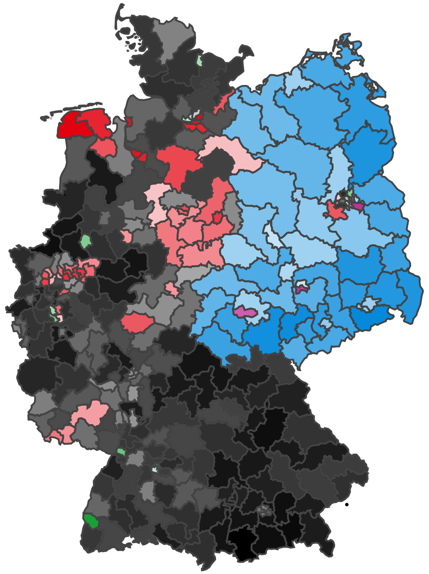
270 out of 299 constituencies correctly predicted (90.3%).

Citizen Forecast
Erststimme Winner



274 out of 299 constituencies correctly predicted (91.6%).

Result
Erststimme Share



Erststimme Winner Forecasts

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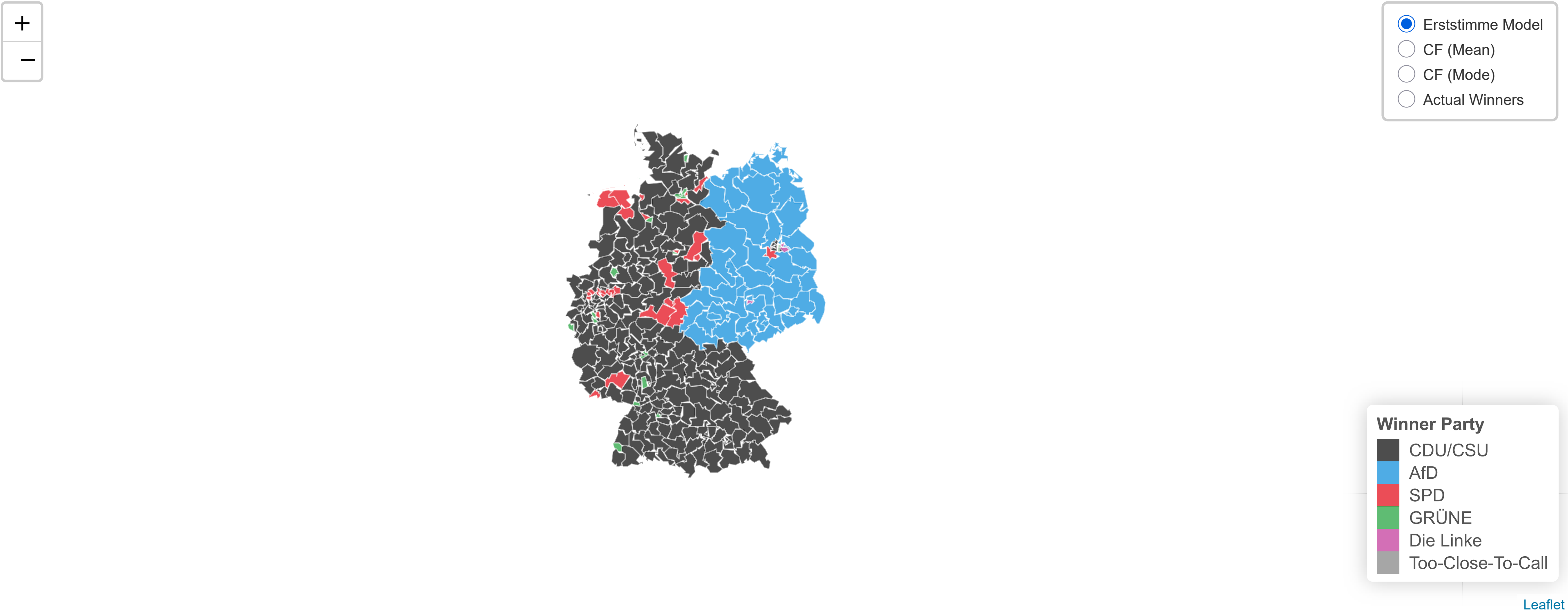

Evaluation: Erststimme Forecasts

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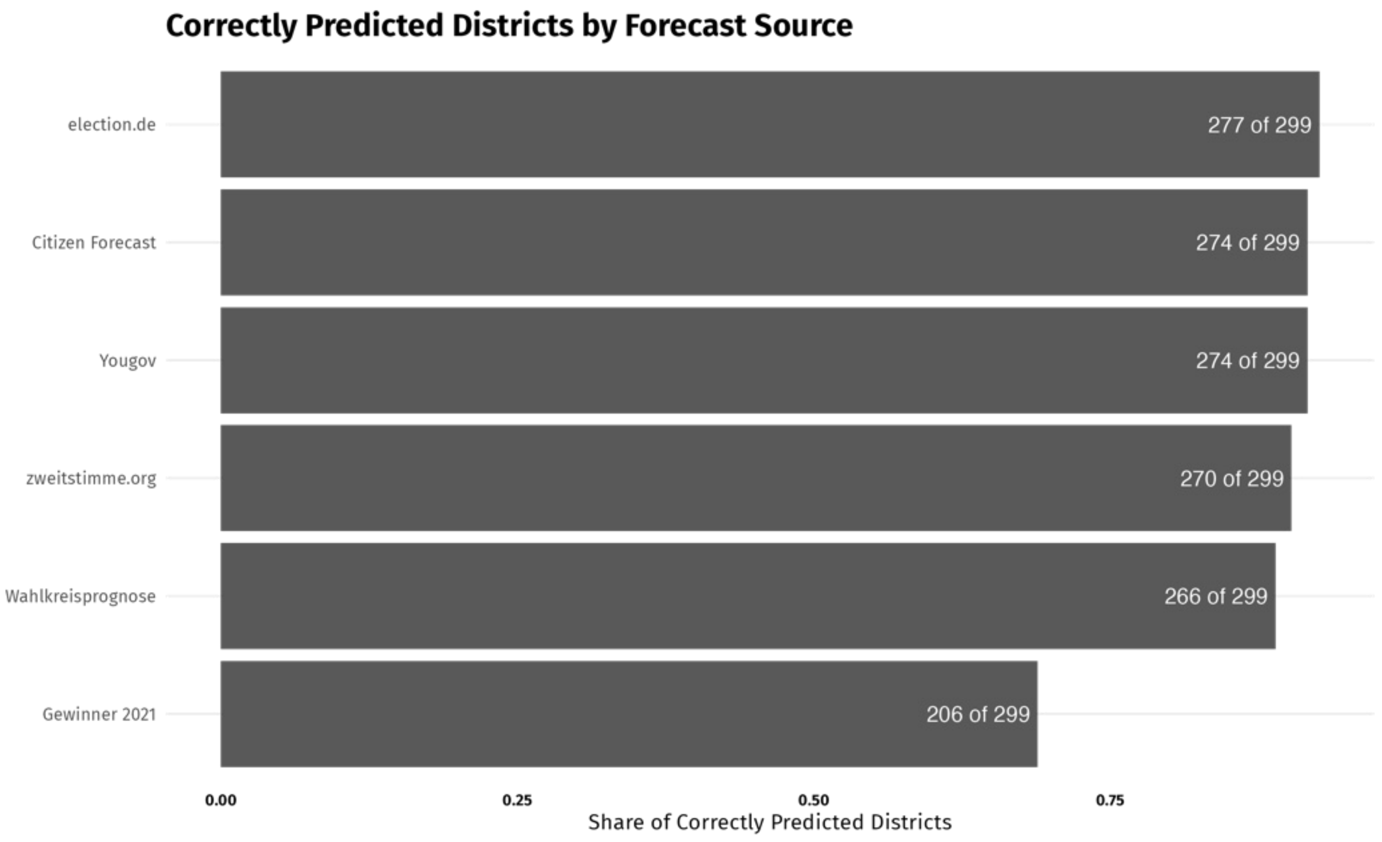
Model Performance

- Erststimme Model: 270 of 299 constituencies correctly predicted (90.3%)
- Citizen Forecast (Mode): 274 of 299 (91.6%)
- Improved performance from previous election

Challenging Areas

- Urban districts more dif cult to predict
- Some unexpected local candidate effects
- Challenges in predicting Left Party strongholds
- Local trends dif cult to capture without district-level data



What did we forecast for Mannheim?

Forecast for Mannheim

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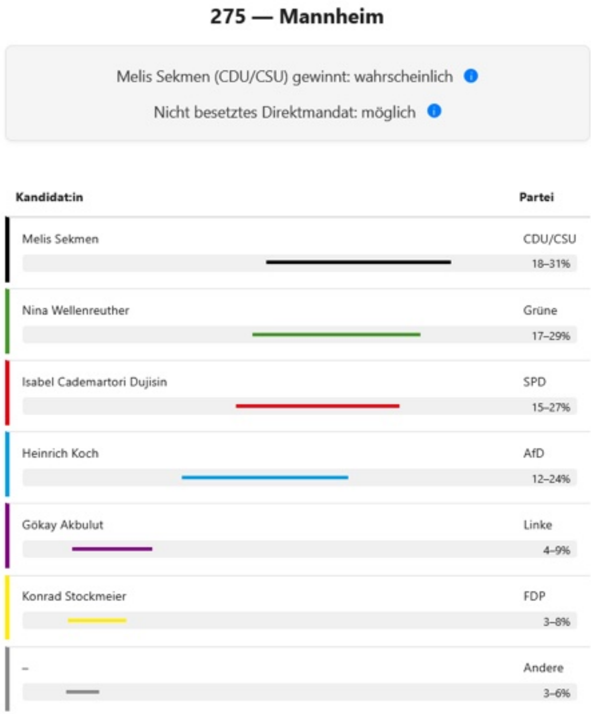
Erststimme Model


275 — Mannheim

Melis Sekmen (CDU/CSU) gewinnt: wahrscheinlich

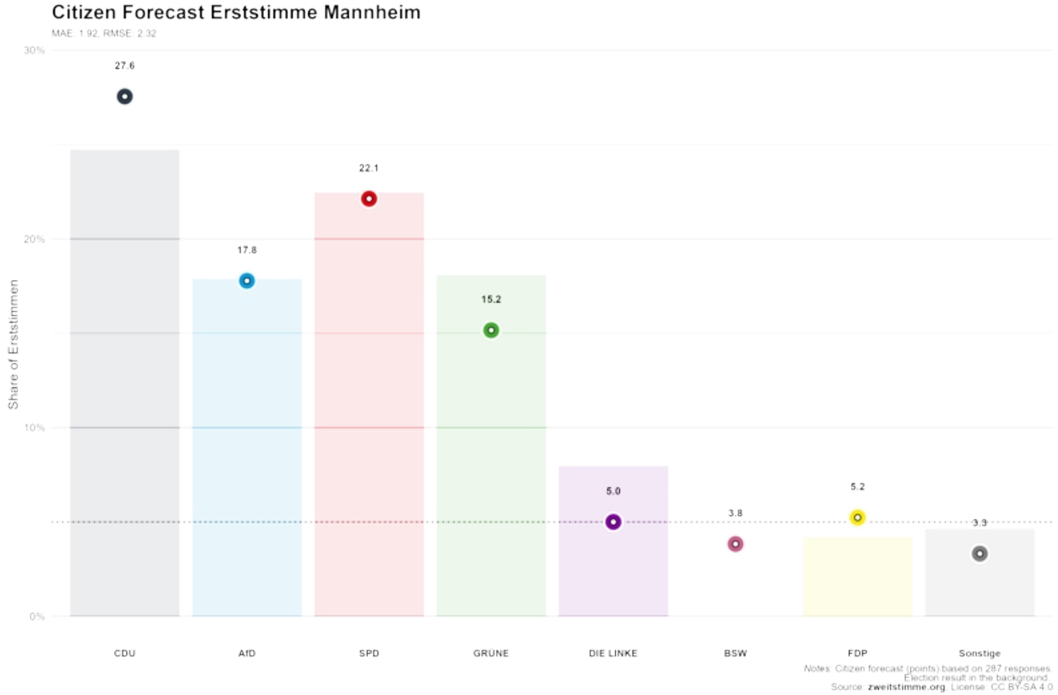
Nicht besetztes Direktmandat: möglich

Kandidat:in	Partei
Melis Sekmen	CDU/CSU 18–31%
Nina Wellenreuther	Grüne 17–29%
Isabel Cademartori Dujisin	SPD 15–27%
Heinrich Koch	AfD 12–24%
Gökay Akbulut	Linke 4–9%
Konrad Stockmeier	FDP 3–8%
–	Andere 3–6%

Citizen Forecast

Winner (Mode): CDU

Vote Share (Mean):



Vacant Districts

Vacant Districts

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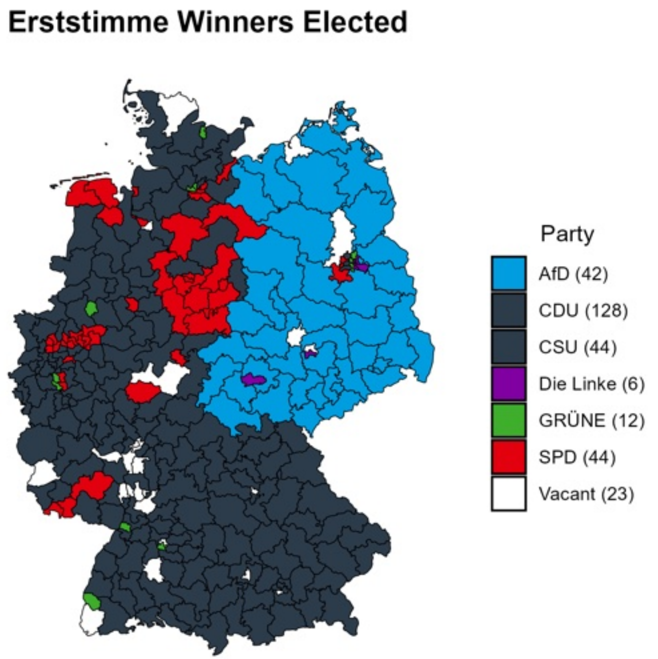
A New Challenge

- First time in German election history
- Winning a district doesn't guarantee a seat
- Parties must obtain suf cient Zweitstimme votes to cover Erststimme seats
- Potential for districts without direct mandates



Estimating Probabilities for Vacant Districts

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Identi cation Process

- Rank district winners by vote share within states
- Compare with party's state-level seat allocation
- Higher probabilities in competitive districts

Methodology

- 10,000 simulations from Zweitstimme model
- Erststimme district predictions for each simulation
- Party inclusion: >5% threshold or >3 district wins
- Seat distribution: 630 Bundestag seats
- State list distribution based on proportional changes and past turnout
- Calculate probability as share of simulations

Forecast for Vacant Districts

Considerations

- Probabilities depend on district win likelihood
- Highest probabilities in competitive districts
- Lower vote shares increase vacancy risk

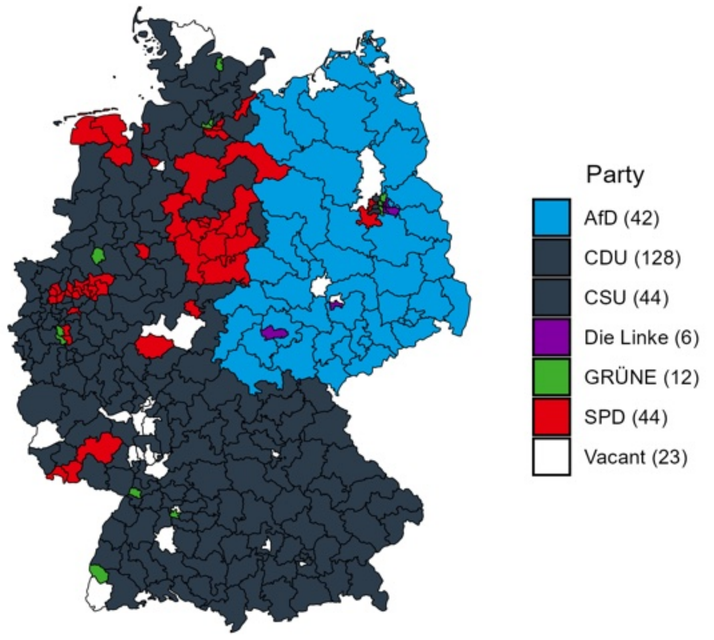
Key Findings

- Most vacant districts were expected for CDU/CSU
- Some AfD districts in Eastern Germany
- Greens and Left Party in city states
- Lower probabilities for SPD districts

Predicting Vacant Districts

Model Forecast
Constituencies without Direct Mandate

Election Result
Constituency Winners

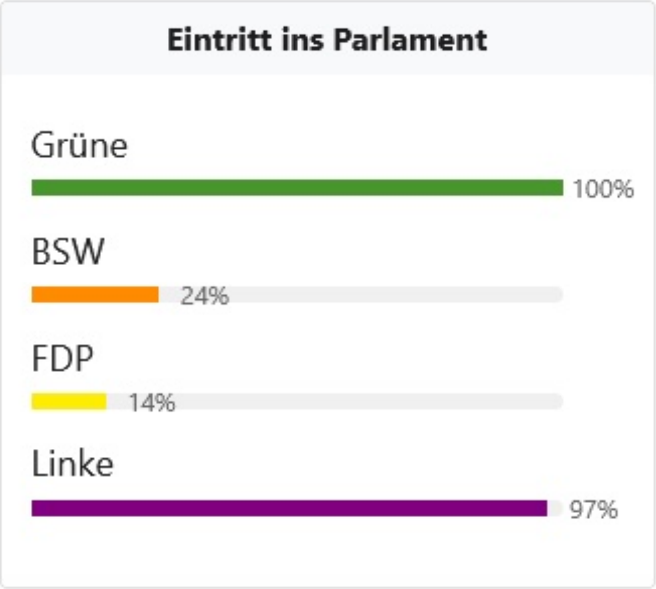


More Forecasts

Entry into Parliament, Coalition and Chancellor

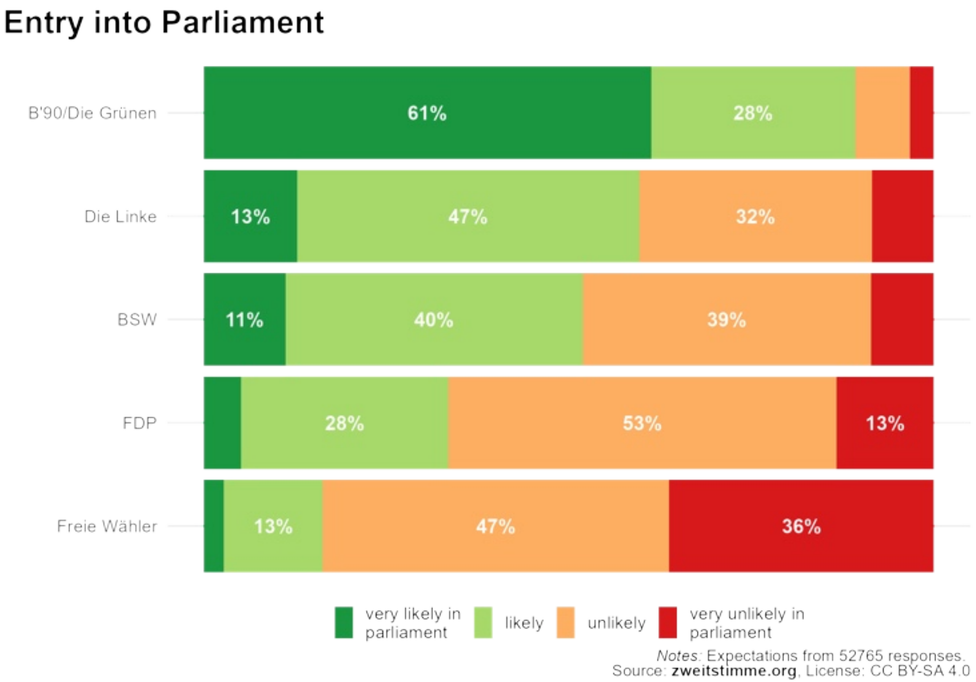
Entry into Parliament

Zweitstimme Forecast



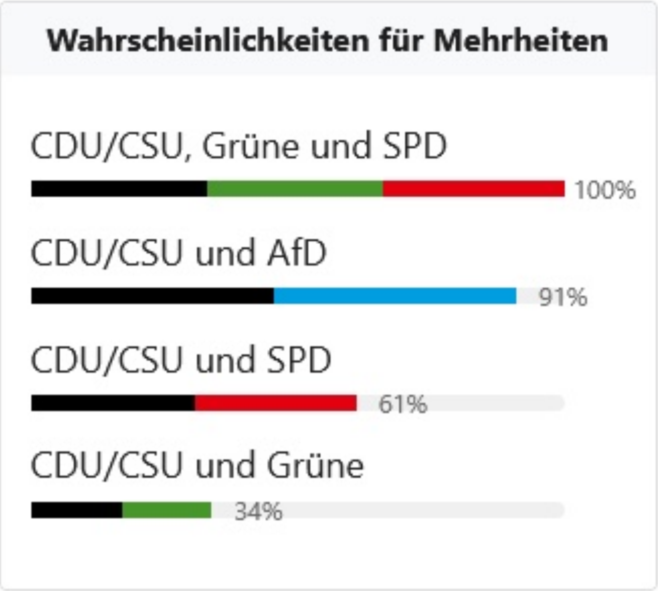
- BSW and FDP correctly predicted to miss 5% threshold
- But they were very close to the threshold

Citizen Forecast



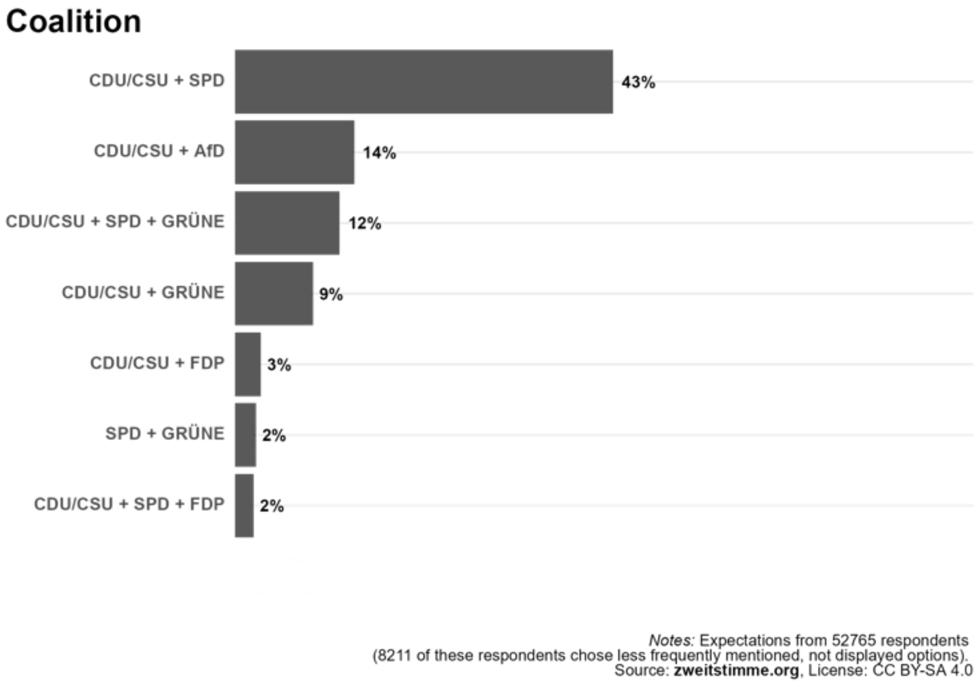
Majorities / Coalition

Zweitstimme Forecast of Majorities



Coming soon: coalition model

Citizen Forecast of Expected Coalition



Chancellor

Citizen Forecast

Conclusion

Conclusion

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Model Performance

- Reliable predictions for both Zweitstimme and Erststimme
- Citizen forecasts also very reliable
- BSW as new party made it more difficult
- Accurate threshold predictions
- Good uncertainty quantification

Future Directions

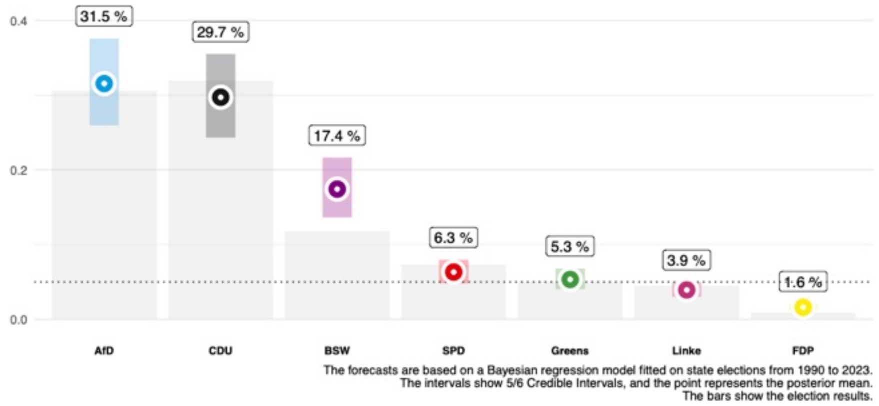
- Combination of citizen forecasting and other models
- Integration of more detailed candidate data
- Enhanced local trend detection
- Improved urban-rural modeling

What else

State elections

- Unique challenges: Different party systems
- Learning between states
- Stoetzer et al. 2025

Forecast Saxony 2024



Coming soon

- Coalition forecasts
- MrP models
- Citizen forecasting: weighted subcrowds
- Experimental evidence: perception and role of polls/forecasts
- Automated pipeline / database for polling data and fundamentals
- Automated forecasts for future state and federal elections